



Resilience Social and Cultural Needs

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ENGCV370

Learning Objectives

- ▶ Understand how resilience strategies impact different demographics of people
- ▶ Recognise the importance of incorporating social and cultural requirements into resilience frameworks

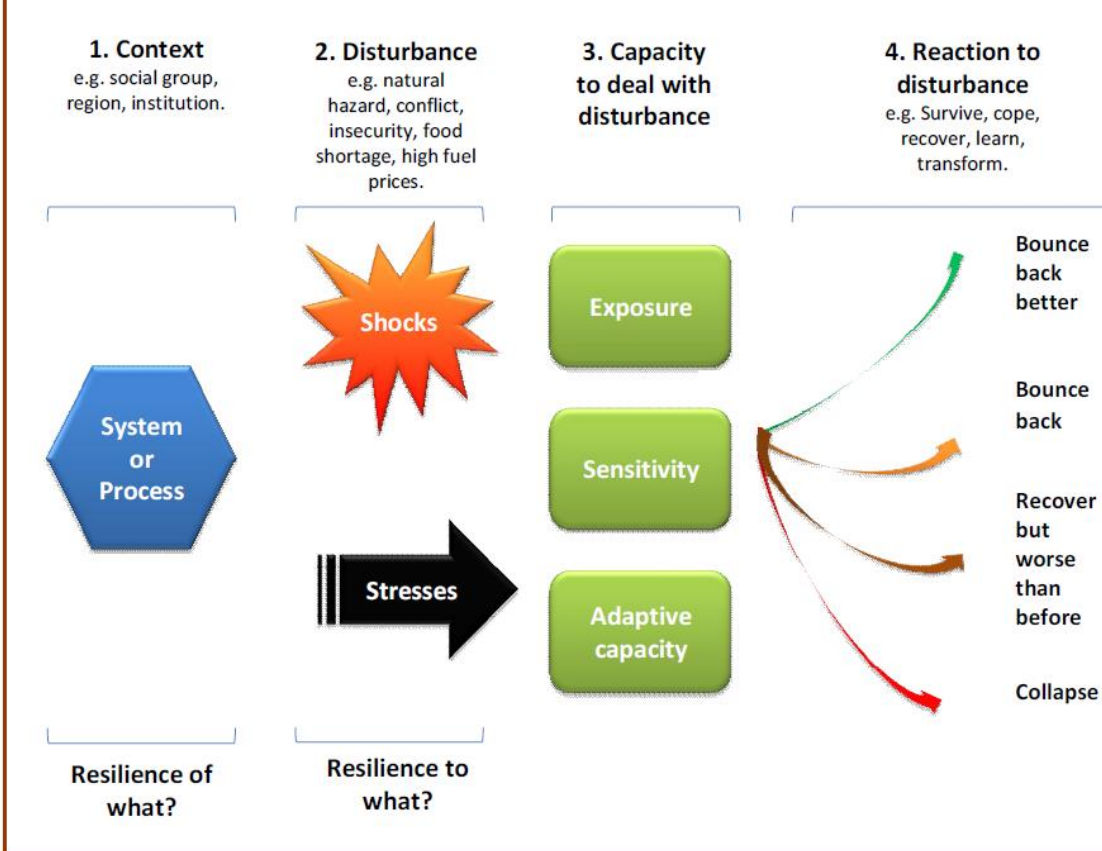
- ▶ I feel that I am generally aware of the impacts of engineering projects on society and the economy?
- ▶ Most large engineering projects have little to no impact on society of communities
- ▶ Most large engineering projects have little to no impact on the economy beyond those employed in design and construction

For the two scenarios presented, briefly respond to the following questions

- A. A large dam is to be constructed in rural area of a developing nation to create a large lake to provide water for irrigation and for generating electricity
 - 1. How do you think such a dam would help the developing nation's economy?
 - 2. How might such a project hurt the economy?
 - 3. How do you think building this large project would benefit the social landscape of the developing nation?
 - 4. How do you think building this large project would (could) hurt the social landscape of the developing nation?
- B. A highway bypass project is proposed around Hamilton
 - 1. How do you think a highway bypass project would help the local economy?
 - 2. How do you think a highway bypass project would hurt the local economy?
 - 3. How do you think a highway bypass project would benefit the social landscape in the community?
 - 4. How do you think a highway bypass project would (could) hurt the social landscape in the community?

Resilience Frameworks – Challenges

The four elements of a resilience framework



Challenges

- ▶ Resilient systems are better able to cope, recover and adapt
- ▶ Fails to account for power relations and complexity of social dynamics
- ▶ Marginalised communities bear the brunt of impacts from natural disasters
- ▶ Lack resources and capacity for successful adaption and mitigation

Key elements for equitable resilience

- ▶ **Recognizing subjectivities**
- ▶ **Inclusion and representation**
- ▶ **Scale**
- ▶ **Transformation**

Recognizing subjectivities

- ▶ How social contexts, power relations and categorization of people determine social and economic entitlements

Inclusion and representation

- ▶ Avoiding processes that disenfranchise some groups

Scale

- ▶ Attending to multiple levels of governance as well as geographical and temporal aspects of scale

Transformation

- ▶ Promoting system(s) **transformation** when existing arrangements degrade well-being or increase risks for certain sections of society

Solutions:

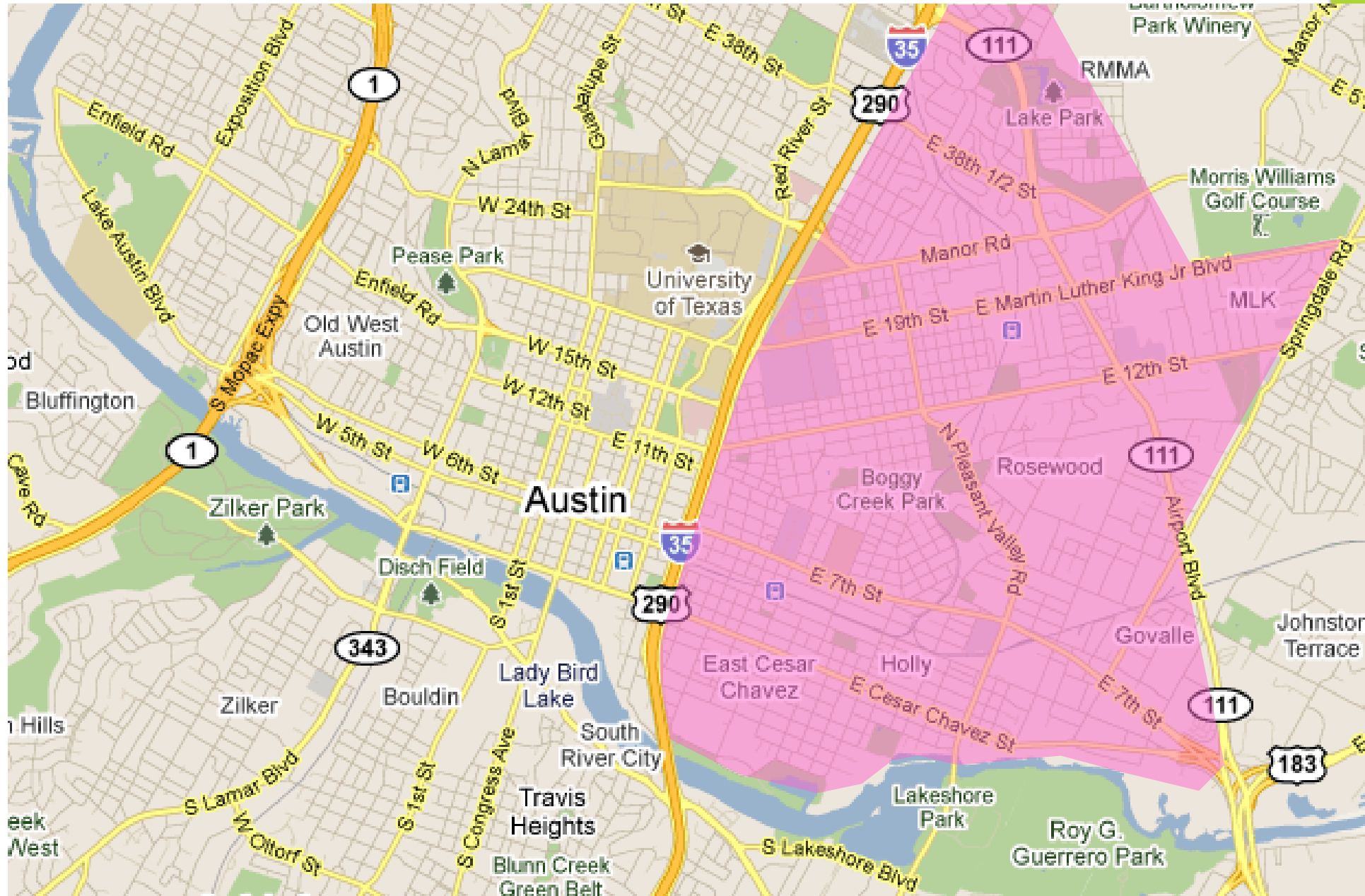
- ▶ Reframing – baseline for disaster readiness
 - ▶ Reinvent – communications
 - ▶ Reinforce – Repetition
 - ▶ Revamp – outreach
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- ▶ Understand the issues and challenges.



Merowe Dam
in Sudan (Nile River)



Austin, TX

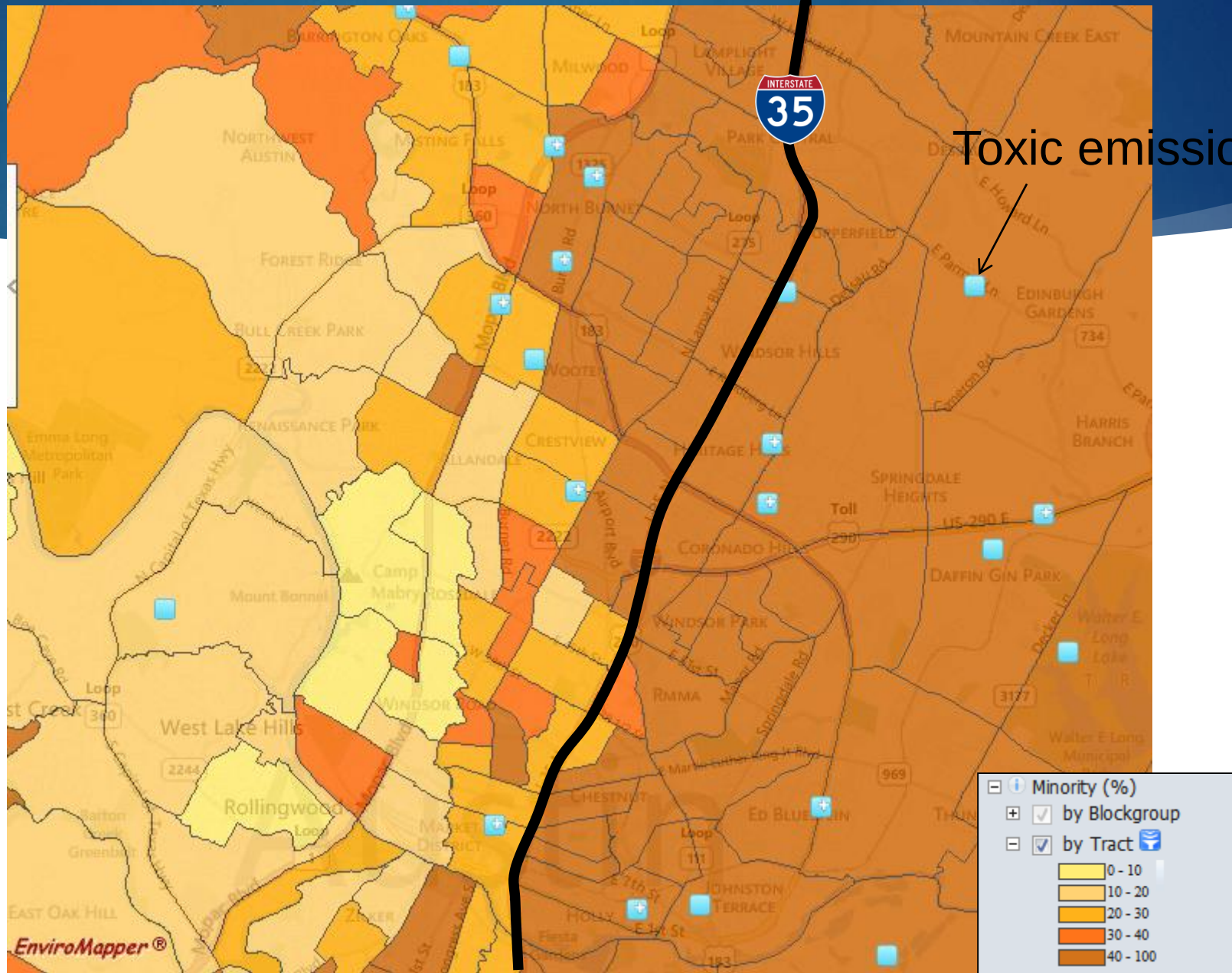






Austin, TX (I-35 segregation)

17









The Study



❖ Cultural Practices

The Study

Kia Niwha Te Ngakau Ki Nga mahi
Arawa, Whakatauki, Purakau

Naku te rourou nau te rourou
ka ora ai te iwi

“Hutia te rito o te harakeke Kei whea to kōmako e kō? Ki mai ki ahau He aha te mea nui o te Ao? Maku e kī atu, he tāngata, he tāngata, he tāngata...”

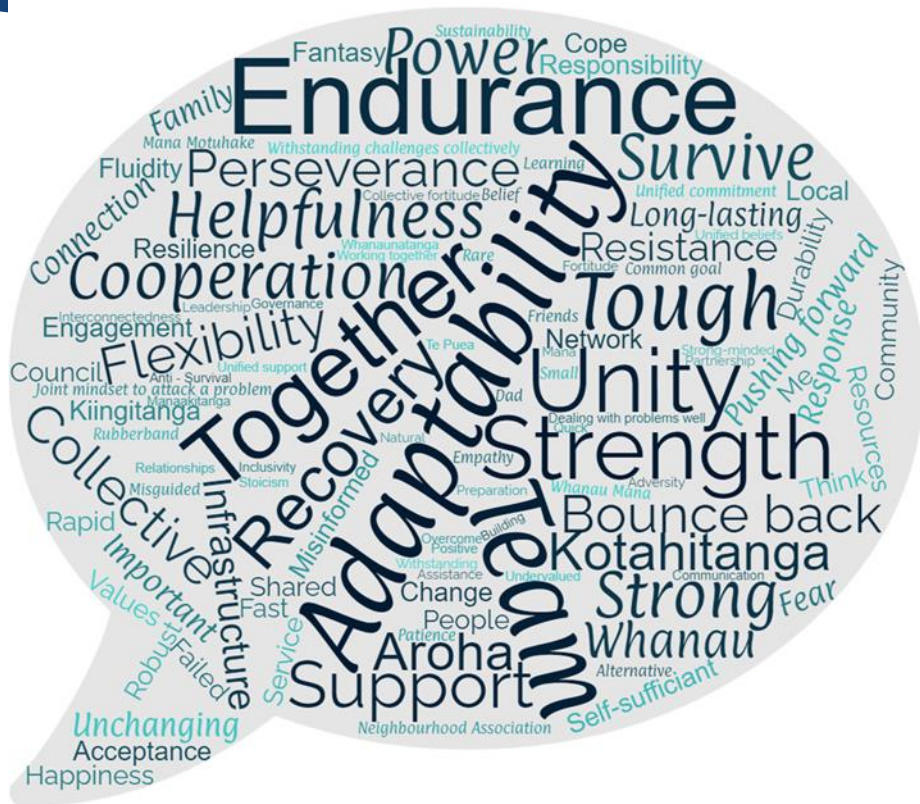
❖ Case Study – 2004 Manawatu Whanganui Flood

The Study



❖ Survey & Interviews

The Study

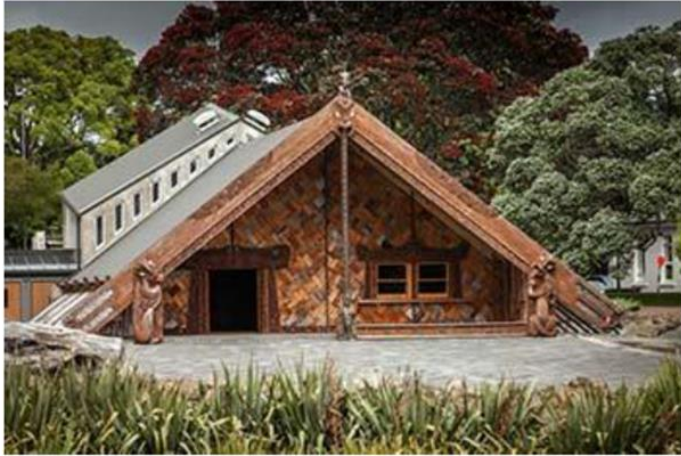


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❖ Survey & Interviews

The Study



Findings

Mātauranga

“Learn and strengthen resources to become resilient. Reflect on appropriate kōrero tuku iho for tīpuna”

Mātauranga
(knowledge)



“Because the wood used to build pā are whole trees, whereas the modern style of wood used is already cut up pieces of a tree, not the whole thing. ‘Together we stand, divided we fall.’”

Findings

Kotahitanga

"Community is central to recovery
post-disaster"

"there is a sense
of community"

Kotahitanga
(unity)



"the wharehenui is not just a mere
bunch of wood and nails
hammered together, it has a
whole spiritual whare inside it. So
double the strength"

Findings

Kaitiakitanga

Iwi work to ensure the well-being of Māori and all others in the region during a disaster – Carter 2018

“It’s got to be the people. Ensuring the safety of the people – look after the whakapapa – check-in with the whakapapa.”

“I feel like our ancestors were more involved and active in each other’s lives. They were less occupied with the world but also they were more of a community so if something happened then they would help each other immediately. “

“Built to resist attacks, famine, disaster in mind”

Kaitiakitanga
(guardianship)



Findings

Whakawhanaungatanga

Whakawhanaunga
(relationships)



“I feel like our ancestors were more involved and actively engaged in each other’s lives. They were less occupied with the world, but also they were more of a community, so if something happened, then they would help each other immediately.”

“Whanau/hapu would come together and work together to reinstate the whare”

Findings

Manākitanga

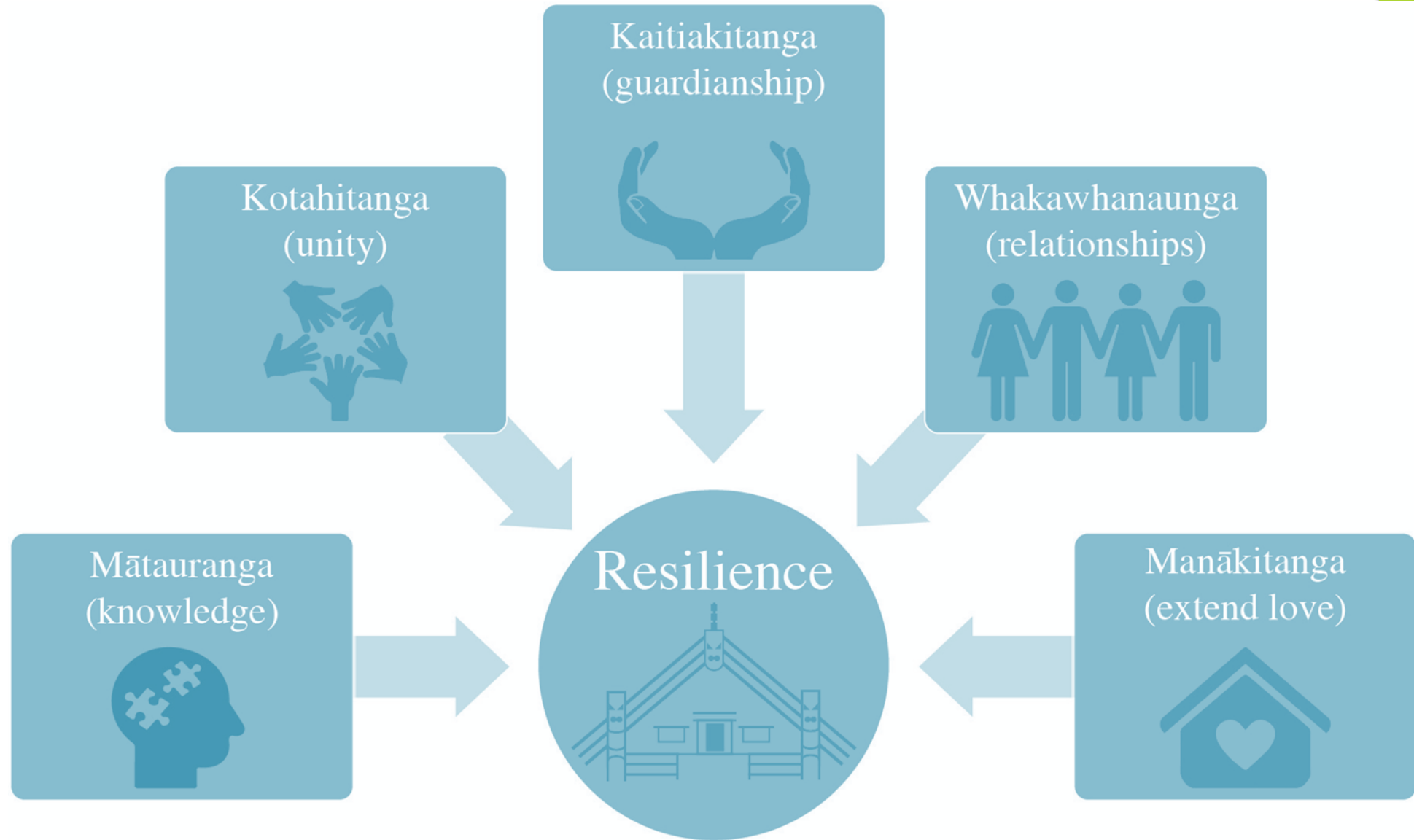
“they would
know one
another and look
after each
other”

Manākitanga
(extend love)



“whanau hapu help each other out,
support each other help raise and
care for the young and old”

“Whare were bare minimum. Communal living means people
were well in touch with each other and their environment.
Connected through whakapapa more often than not -
something more to fight and care for.”



Takeaways



Nepal 2015 Earthquake
PC: Sonny Inbaraj.

Takeaways



Photo: Taneesha Mohan, SEI

Takeaways



Photo: DANIEL BEREHULAK/THE NEW YORK TIMES/REDUX

Social Vulnerabilities